DA14 Exploring distributions with seaborn

Week 5, Explore with NumPy and seaborn | Session 14 of 24 | Python for Data Analysis | New library: seaborn

Today's question. What does a typical sale look like, and does it differ between the three shops?

Objective. By the end: you can draw a histogram and a box plot with seaborn and describe the shape, the middle and the outliers of a column.

> **WHERE THIS IS USED**
> A histogram shows what a typical day looks like, and a box plot shows how groups differ. Club analysts compare players and teams with plots like these. Which two shops would you compare first?

Before this session: DA13 NumPy: the numbers underneath pandas. If you cannot explain the last example, use its error card with the instructor first.

Words for today

Word	What it means	Think of it like this
seaborn	A library that draws statistical charts. It sits on top of matplotlib and takes a DataFrame directly.	A set of ready-made stencils for charts.
exploratory analysis	Looking at the data from several sides to see what is there before you answer a question.	Walking round a shop before you count the stock.
distribution	How the values of one column are spread out.	How the pupils spread across height bands.
box plot	A box from the lower to the upper quartile with a line at the median and dots for outliers.	A five-number summary drawn as a picture.
skewed	Bunched at one end with a long tail at the other.	Most pupils short and a few very tall.
group comparison	Drawing the same chart for each group side by side.	Comparing the three shops on one page.

Block A. A histogram of sale values

sns.histplot draws a histogram from a DataFrame. Name the DataFrame in data= and the column in x=. A map turns branch ids into names.

Predict. Predict whether the histogram is balanced or skewed to the right. Write your guess on paper first.

Your guess, written before you run it

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
sunrise = pd.read_csv("sunrise_sales.csv")
sunrise["line_value"] = sunrise["units"] * sunrise["unit_price_ugx"]
sunrise["branch"] = sunrise["branch_id"].map({1: "Central Market", 2: "Campus Gate", 3:
    "Roadside"})
sns.histplot(data=sunrise, x="line_value", bins=30)
plt.xlabel("Value of one sale (UGX)")
plt.show()
print(round(sunrise["line_value"].skew(), 1))
```

Run it, then compare. The histogram is bunched at the low end with a long tail to the right. The skew is 7.3. A positive skew above 1 means a long right tail. Most sales are small and a few are large.

Investigate.

1. What does the long right tail mean for the mean and the median?
2. What does exploratory analysis do that a single total does not?

Modify. Draw the histogram again with bins=60 and say what changes.

Block B. A box plot for each shop

sns.boxplot draws one box for each group. The line inside the box is the median. The box runs from the lower quartile to the upper quartile. Dots beyond the whiskers are outliers.

Predict. Predict which shop has the lowest median sale. Write your guess on paper first.

Your guess, written before you run it

```
sns.boxplot(data=sunrise, x="branch", y="line_value")
plt.ylabel("Value of one sale (UGX)")
plt.show()
medians = sunrise.groupby("branch")["line_value"].median()
print(medians)
```

Run it, then compare. The medians are 2500 at Central Market, 2000 at Campus Gate and 2500 at Roadside. The boxes look alike and every shop has dots well above its whiskers: bulk sales happen in all three. The shops differ more in how many sales they make than in how big a typical sale is.

Investigate.

1. Do the shops differ much in typical sale value?
2. What do the dots above each box show?

Modify. Use the box plot to compare the units column across shops. Print the median units for each shop.

Make

Explore the daily customer counts. Draw a histogram of customers_counted and a box plot of customers_counted for each branch. Write the median for each branch and one finding in a sentence.

Save your work as DA14. Keep the worked example and put your own attempt below it. Write two sentences: what you changed and how you checked it.

Error card

You see	It means	Fix
NameError: name 'sns' is not defined	seaborn was not imported in this session.	Run import seaborn as sns in the first cell, then run the chart cell.
ValueError: Could not interpret value for x	The column name in x= or y= is not in the DataFrame.	Print list(daily.columns) and copy the exact name.

Exit ticket

A histogram shows the shape of one column and a box plot lets me compare groups.

AI use follows AI_USE_POLICY.pdf. In weeks 1 to 6 no AI-generated solutions are used for exercises, labs or build tasks. Practise for 20 to 30 minutes between classes by repeating the Make task with one value changed. If you are stuck for 30 minutes, save the error and bring it to class. A file saved on your laptop has not been shared until you use the route in START_HERE.pdf.